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"""
Epic 1: Data Collection
Project: Rising Waters - Flood Prediction
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Downloads the flood prediction (rainfall) dataset from Kaggle
and loads it into a pandas DataFrame for use in later epics.

Dataset: https://www.kaggle.com/arbethi/rainfall-dataset

Requirements:
    pip install kagglehub pandas openpyxl

Usage:
    python epic1_data_collection.py
"""

import os
import sys
import pandas as pd

def download_dataset():
    """Download the rainfall dataset from Kaggle using kagglehub."""
    try:
        import kagglehub
    except ImportError:
        sys.exit(
            "Missing dependency 'kagglehub'.\n"
            "Install it with: pip install kagglehub"
        )

    print("Downloading dataset from Kaggle...")
    path = kagglehub.dataset_download("arbethi/rainfall-dataset")
    print(f"Dataset downloaded to: {path}")
    return path

def load_dataset(folder_path):
    """Find and load the dataset file (xlsx or csv) into a DataFrame."""
    files = [
        f for f in os.listdir(folder_path)
        if f.lower().endswith((".xlsx", ".csv"))
    ]

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if not files:
    raise FileNotFoundError(
        f"No .xlsx or .csv file found in {folder_path}. "
        f"Files present: {os.listdir(folder_path)}"
    )

file_path = os.path.join(folder_path, files[0])
print(f"Loading file: {file_path}")

if file_path.lower().endswith(".xlsx"):
    df = pd.read_excel(file_path)
else:
    df = pd.read_csv(file_path)

return df

def main():
    folder_path = download_dataset()
    df = load_dataset(folder_path)

    print("\n--- Dataset Summary ---")
    print("Shape:", df.shape)
    print("\nColumns:", df.columns.tolist())
    print("\nFirst 5 rows:")
    print(df.head())

    # Save a local copy for reuse in later epics (EDA, preprocessing, etc.)
    output_dir = "data"
    os.makedirs(output_dir, exist_ok=True)
    output_path = os.path.join(output_dir, "rainfall_dataset.csv")
    df.to_csv(output_path, index=False)
    print(f"\nSaved local copy to: {output_path}")

if __name__ == "__main__":
    main()

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